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SCHOOL OF COMPUTING

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

SCSA3012- KNOWLEDGE MANAGEMENT SYSTEMS

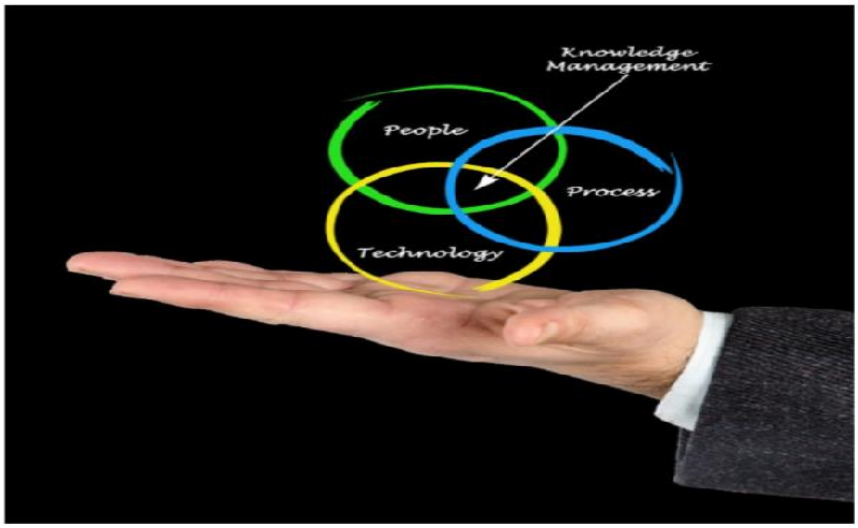
UNIT 1 KNOWLEDGE MANAGEMENT 9Hrs

KM Myths–KM Life Cycle–Understanding Knowledge–Knowledge, intelligence –Experience – Common Sense – Cognition and KM – Types of Knowledge – Expert Knowledge – Human Thinking and Learning.

KNOWLEDGE MANAGEMENT

Knowledge management is the systematic management of an organization's knowledge assets for creating value and meeting tactical & strategic requirements. It consists of the initiatives, processes, strategies, and systems that sustain and enhance the storage, assessment, sharing, refinement, and creation of knowledge.

Each enterprise should define knowledge management in terms of its own business objectives. Knowledge management is all about applying knowledge in new, previously overburdened or novel situations.



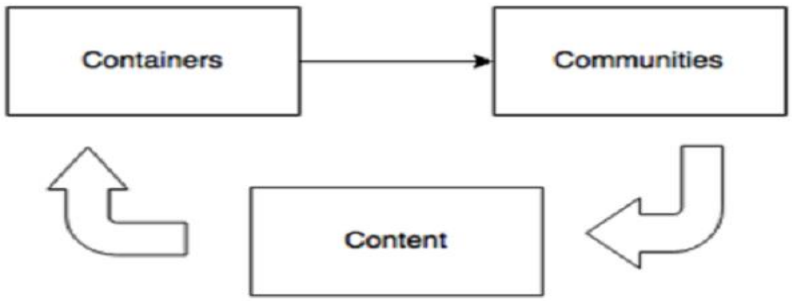
Knowledge Management is a Continuous Cycle. Knowledge management is currently seen as a continuous cycle of three processes, namely −

- Knowledge creation and improvement
- Knowledge distribution and circulation
- Knowledge addition and application

Knowledge management expresses a deliberate, systematic and synchronized approach to ensure the full utilization of the company's knowledge base, paired with the potential of individual skills, competencies, thoughts, innovations, and ideas to create a more efficient and effective company.In simple words, knowledge management incorporates both **holding and storing** of the knowledge perspective, with respect to the intellectual assets.It is the deliberate and systematic collaboration of an organization's people, technology, processes, style and structure in order to add value through reuse and innovation.

KNOWLEDGE MANAGEMENT THEORY

There are three distinct perspectives on Knowledge Management which leads to a different estimation and a different definition.



Knowledge management is a business activity with two primary aspects −

- Executing the knowledge component of business activities as an explicit concern of business in strategy, policy, and practice at all levels of the organization.
- Maintaining a direct link between an organization's intellectual assets both explicit (recorded) and tacit (personal know-how) and positive business results.

What Cognitive Science or Knowledge Science Perspective Says?

Knowledge management is the transformation of knowledge in the form of insights, understandings, and practical know-how that we all possess in other manifestations like books, technology, practices, and traditions within organizations of all kinds and in society in general.

According to the Process/Technology Perspective

Knowledge management is the concept under which information is changed into actionable knowledge and made available effortlessly in a usable form to the people who can leverage it according to their needs.

Why Knowledge Management?

Application of Knowledge Management (KM) lie in the below four key areas

- **Globalization of Business** − Organizations today are more universal i.e., they are operating in multiple sites, multilingual, and multicultural in nature.
- **Leaner Organizations** − Organizations are adopting to a lean strategy where they understand customer value and focus on key processes to continuously increase it. The ultimate goal is to provide perfect value to the customer through a perfect value creation process that has zero waste.
- **Corporate Amnesia** − We are freer as a workforce, which creates issues regarding knowledge continuity for the organization and places with continuous learning demands from knowledge worker. We no longer expect to spend our entire work life with the same organization.
- **Technological Advances** − The world is more connected with the advent of websites, smart phones and other latest gadgets. Advancements in technology has not only helped in better connectivity but also changed expectations. Companies are expected to have online presence round the clock providing required information as per the customer needs.

Knowledge Management serves as one of the major response to the challenge of trying to handle this complex, information overloaded work environment. As such, Knowledge management is perhaps best clustered as a science of complexity.

KNOWLEDGE MYTHS

1. Knowledge management is an end unto itself.

In fact, while creating new knowledge, organizing it so others can use it and passing it on to succeeding generations is a defining human trait, business organizations should use knowledge management to build market value and competitive strength.

2. Knowledge management is just for professional services firms and other "intellectual" businesses.

Not so. Knowledge is a big percentage of the value of all goods and services produced in every industry.

3. Knowledge management just means hiring smart people.

While it is true that people are the main sources of knowledge, it's what people do, not just what they know, that adds value to firms.

4. Knowledge management means implementing expensive technology.

Technology certainly plays a key role, especially in distributing knowledge. But technology alone won't improve a company's knowledge management or make it more competitive.

5. Knowledge management means creating huge, unwieldy databases.

It's true that many firms follow the "Grandma's attic" approach to saving any information that might possibly be useful someday. But these aren't the firms that get business value from their knowledge. Knowledge yields value when your people know where it is, know how to get at it; know it will help them; and join in keeping it current, practical and useful.

6. Knowledge management is a "Field of Dreams" -- just build it, and they will come.

In the movie of the same name, a mystical voice tells the main character that if he builds a baseball field in his backyard, the legends of baseball will appear. That voice won't help you with knowledge management. Building a new technology or process for creating and applying knowledge isn't enough to inspire participation in knowledge management or yield business results.

7. Good knowledge management is driven by a good chief knowledge officer or chief learning officer.

The creation, distribution and application of knowledge drives the value of an organization's goods and services and determines its market value. Whether you recognize and cultivate them or not, your knowledge processes are at the heart of your business. They can't be left in the hands of one executive, however effective.

8. Knowledge management is just for Americans.

Because the marketplace is global, the creation, distribution and application of knowledge must also be global and not confined to one region, language or culture.

9. Knowledge management isn't like other good management practices and processes.

Managing knowledge does not mean reinventing management. The elements of effective financial management, such as building theoretical knowledge, developing a shared vocabulary and implementing best practices and processes, also drive effective knowledge management.

10. Knowledge management is a fad.

The term "knowledge management" might fade away or be replaced. But the enormous contribution of knowledge management to business value will only grow and the activities that create value from knowledge will always be vital to success.

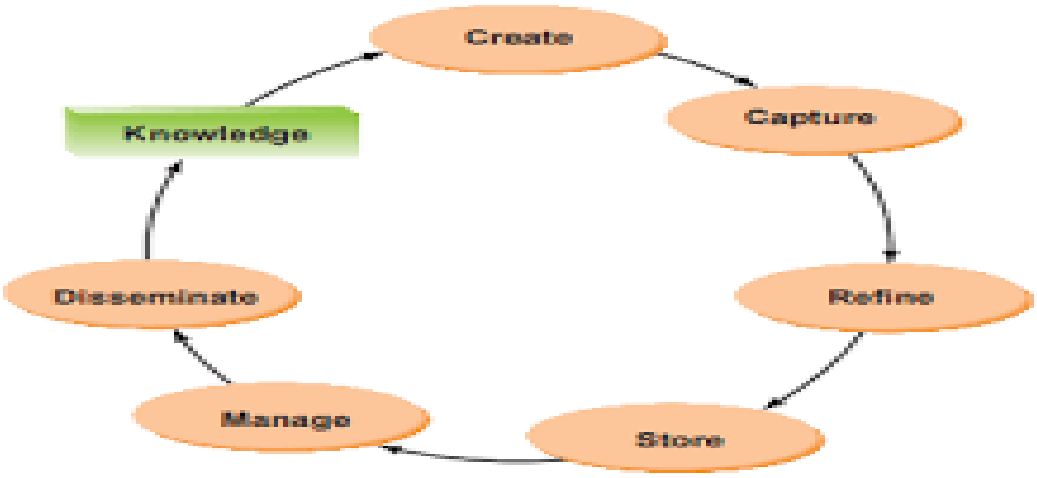
KNOWLEDGE MANAGEMENT PROCESS

the knowledge management process involves knowledge acquisition, creation, refinement, storage, transfer, sharing and utilization. This process can be synthesized this a little further. Effective knowledge management system typically goes through three main steps:

1. **Knowledge Creation:** During this step, organizations identify and document any existing or new knowledge that they want to circulate across the company.
2. **Knowledge Storage:** During this stage, an information technology system is typically used to host organizational knowledge for distribution. Information may need to be formatted in a particular way to meet the requirements of that repository.
3. **Knowledge Sharing:** In this final stage, processes to share knowledge are communicated broadly across the organization. The rate in which information spreads will vary depending on organizational culture. Companies that encourage and reward this behavior will certainly have a competitive advantage over other ones in their industry.

KNOWLEDGE MANAGEMENT LIFE CYCLE

Knowledge management cycle is a process of transforming information into knowledge within an organization. It explains how knowledge is captured, processed, and distributed in an organization. In this chapter, we will discuss the prominent models of knowledge management cycle.



The knowledge management system cycle.

Till date, four models have been selected based on their ability to meet the growing demands. The four models are the Zack, from Meyer and Zack (1996), the Bukowitz and Williams (2000), the McElroy (2003), and the Wiig (1993) KM cycles.

Zack	Bukowitz& Williams	WIIG	McElroy
Acquisition	Get	Creation	Learning
Refinement	Use	Sourcing	Validation
Store	Learn	Compilation	Acquisition
Distribution	Contribute	Transformation	Integration
Presentation	Assess	Application	Completion

[1]Zack Knowledge Management Model

The Zack model is extracted from work on the design and development of information products. In Meyer and Zack's approach, the network between each stage is designed to be logical and standardized.

In this cycle, the major developmental stages of a knowledge repository are analyzed and mapped to the stages of a KM cycle.

The stages are acquisition, refinement, storage/retrieval, distribution, and presentation/use. This cycle is also known as the "refinery."

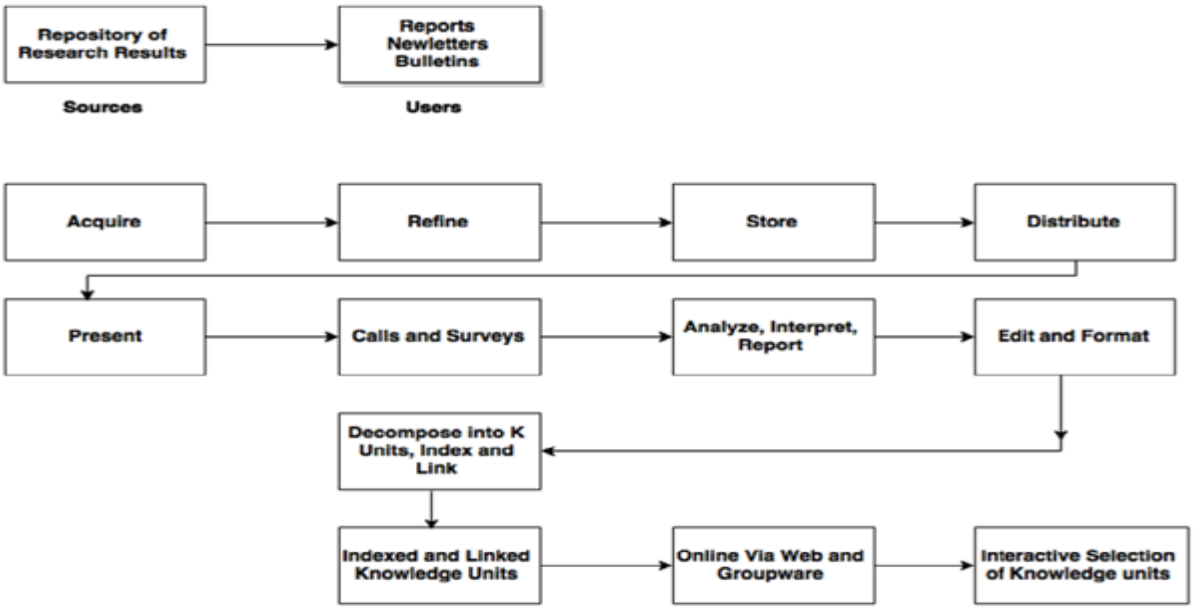
Acquisition of Data or Information

Acquisition deals with issues regarding origin of raw materials such as scope, breadth, depth, credibility, accuracy, timeliness, relevance, cost, control, and exclusivity.

The guiding principle is the well- known proverb of "garbage in, garbage out." That is, highest quality source data is required, else the intellectual products produced downstream will be lower.

Refinement

Refinement may be physical (like migrating from one medium to another) or logical (like restructuring, relabeling, indexing, and integrating.)



Refining also defines cleaning up (like sanitizing content so as to ensure complete anonymity of sources and key players involved) or standardizing (like conforming to templates of a best practice or lessons learned as used within that particular organization).

This stage also adds up to the value by creating more readily usable knowledge objects and by storing the content more flexibly for future use.

Storage / Retrieval

Storage or Retrieval forms a bridge between the upstream addition and refinement stages that feed the repository and downstream stages of product generation. Storage can be physical (file folders, printed information) as well as digital (database, knowledge management software).

Distribution

Distribution defines how the product is to be delivered to the end-user (like fax, print, email) and encloses not only the medium of delivery but also its timing, frequency, form, language, and so on.

Presentation

Context plays an important role in Presentation or Application stage. The performance of each of the preceding value-added steps is evaluated here – for example, does the user have enough context to be able to make use of this content? If not, the KM cycle has failed to deliver value to the individual and ultimately to the company.

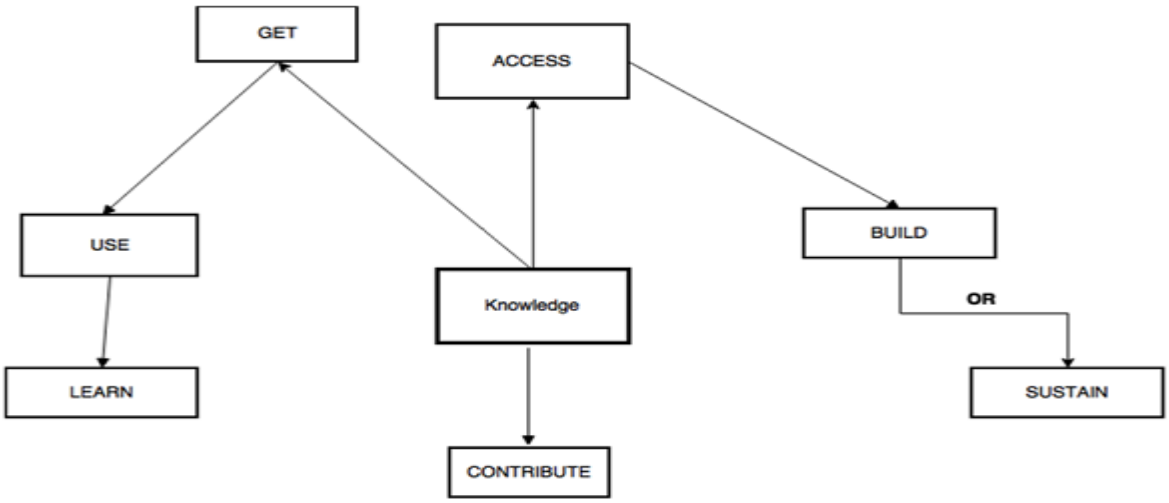
The repository and the "refinery" combined enable the management of valuable knowledge of a firm. In this cycle, there is also an impression of having to continually renew the repository and the refinery in order to avoid elimination.

The Meyer and Zack model is one of the most complete picture of the key elements engaged in the knowledge management model. To be specific the notion of refinement is a crucial stage in the KM cycle and one that is often neglected.

[2]Bukowitz & Williams Model

Bukowitz and Williams portray a knowledge management process framework that outlines "how organizations generate, maintain and expand a strategically correct stock of knowledge to create value".

In this framework, knowledge includes knowledge repositories, relationships, information technologies, communications infrastructure, functional skill sets, process know-how, environmental responsiveness, organizational intelligence, and external sources.



These stages aim on more long-range processes of matching intellectual capital to strategic needs.

- **Get Stage** is the first stage, it consists of seeking out information required in order to make decisions, solve problems, or innovate.
- **Use Stage** is the next stage, and it deals with how to combine information in new and interesting ways in order to foster organizational innovation. The spotlight is primarily on individuals and then on groups.
- **The Learn Stage** points to the formal process of learning from experiences as a means of creating competitive gain. Learning in enterprises is important because it serves the transition step between the application of ideas and the generation of new ones.
- **The Contribute Stage** of the Knowledge Management cycle deals with encouraging employees to post what they have learnt to the communal knowledge base (like a repository). Only in this way, can individual knowledge be made visible and available to the entire organization, where and when appropriate.

[3]McElroy Model

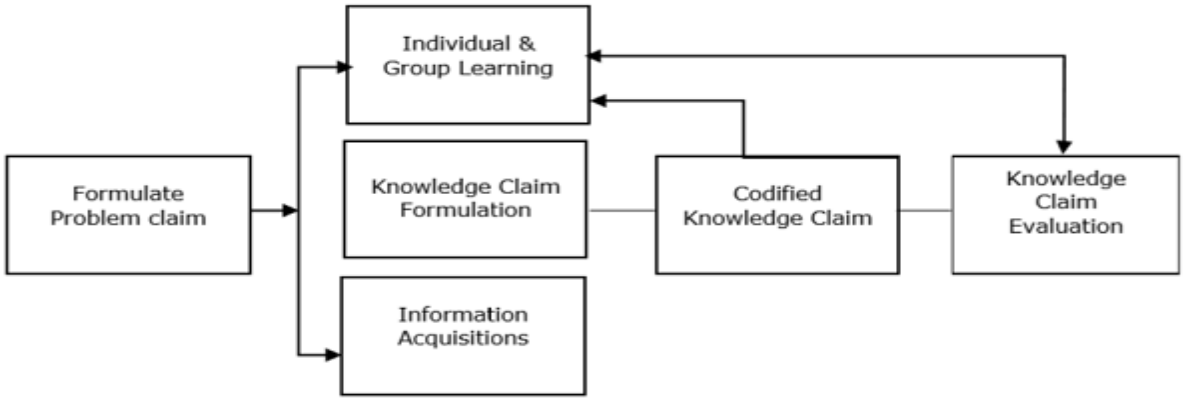
McElroy outlines a knowledge life cycle that consists of the processes of knowledge production and knowledge integration, with a series of feedback loops to organizational memory, beliefs, and claims and the business-processing environment.

- Problem claim formulation is an attempt to learn and state the specific nature of the detected knowledge gap.
- Knowledge claim formulation acts as a response to approved problem claims via information acquisition and individual and group learning.
- New knowledge claims are tested and examined through knowledge claim evaluation processes.

- Evaluation of knowledge claims results in surviving knowledge claims that will be integrated as new organizational knowledge or falsified/undecided knowledge claims.

Experience gained from the application of knowledge in the organizational knowledge base leads to new claims and resulting beliefs, triggering the cycle to begin all over again.

In knowledge production, the primary processes are individual and group learning. Knowledge claim formulation, information acquisition; codified knowledge claim and knowledge claim evaluation.



These knowledge production processes can be briefed as −

- Individual and group learning marks the first step in organizational learning.
- Knowledge claim validation includes codification at an organizational level.
- A formalized procedure is essential for the receipt and codification of individual and group innovations.
- Information addition is the process by which an organization deliberately or serendipitously acquires knowledge claims or information produced by others, usually external to the company. This stage plays a basic role in formulating new knowledge claims at the organizational level.

Knowledge integration is the process by which an organization announces new knowledge claims to its operating environment and retires old ones. It includes all knowledge transmission such as teaching, knowledge sharing, and other social activities that either connects an understanding of previously produced organizational knowledge to knowledge workers or accommodate newly minted knowledge.

One of the advantages of the McElroy cycle is the clear description of how knowledge is examined and a conscious decision is made as to whether or not it will be included into the organizational memory. The authorization of knowledge is a step that clearly differentiates knowledge management from document management. The KM cycle aims at processes to identify knowledge content that is of value to the organization and its employees

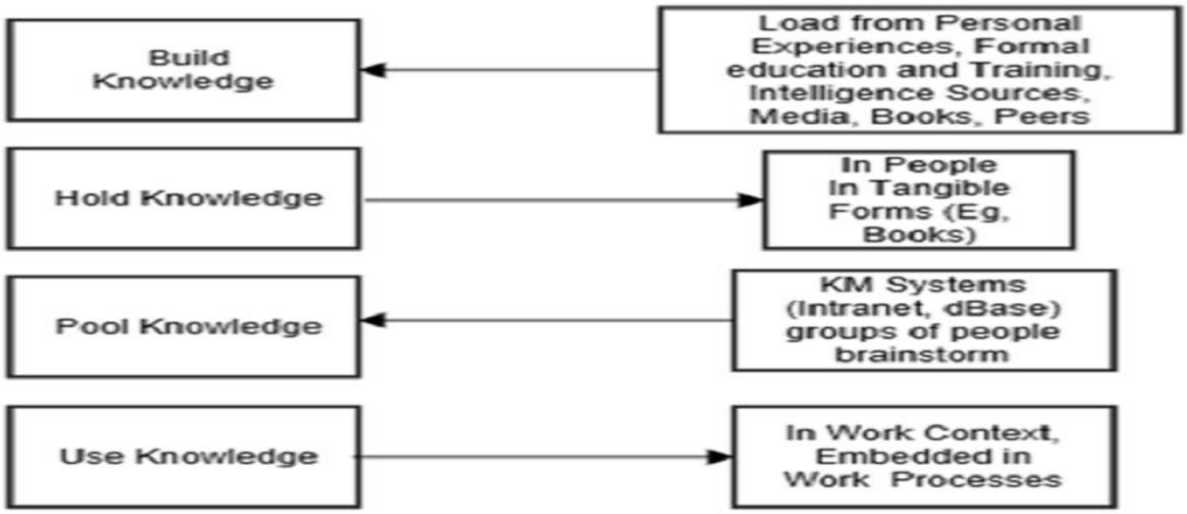
[4]WIIG Model

WIIG highlights the three conditions that need to be present for an organization to conduct its business successfully.

- It must have a business (commodities/services) and customers.
- It must have resources (people, budget, and facilities).
- It must have the strength to act.

WIIG marks the major purpose of KM as an effort "to make the organization intelligent-acting by facilitating the creation, accumulation, deployment and use of quality knowledge." WIIG's KM cycle shows how knowledge is built and used as individuals or as organizations.

The following figure shows the four major steps of the WIIG model.

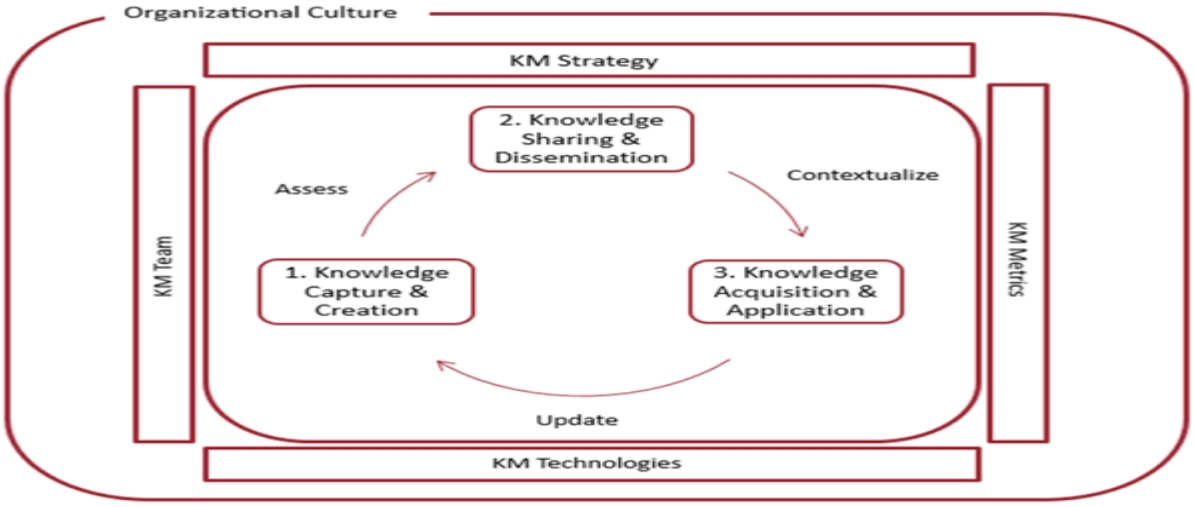


- **Building knowledge** − From external and internal knowledge sources
- **Holding knowledge** − Storing the information in a particular form
- **Pooling knowledge** − Through intranets and knowledge management portals
- **Applying knowledge** − In the context of work embedded in process

Integrated Cycle

The following are the three major stages of integrated cycles of knowledge management strategy when introduced in any organization −

- Knowledge capture and/or creation
- Knowledge sharing and dissemination
- Knowledge acquisition and application



Knowledge Capture states the identification and frequent codification of existing (usually previously unnoticed) internal knowledge and know-how within the organization and/or external knowledge from the environment.

Knowledge Creation is the advancement of new knowledge and know-how innovations that did not have a previous existence within the organization.

Once, it is clear that the newly identified content is of sufficient value, the next step is to contextualize this content. This involves maintaining a connection between the knowledge and those knowledgeable about that content.

Contextualization also indicates identifying the key elements of the content in order to better match to a variety of users. Finally, contextualization succeeds to when the new content is firmly, precisely yet seamlessly, embedded in the business processes of the enterprise.

The knowledge management cycle is then rechecked as users understand and decide to make use of content.

The users will update usefulness, and they will signal when it becomes out of date or when this knowledge is not applicable.

UNDERSTANDING KNOWLEDGE

Knowledge can be defined as the ``understanding obtained through the process of experience or appropriate study." Knowledge can also be an accumulation of facts, procedural rules, or heuristics. A ***procedural rule*** is a rule that describes a sequence of actions. A ***fact*** is generally a statement representing truth about a subject matter or domain. A ***heuristic*** is a rule of thumb based on years of experience. ***Intelligence*** implies the capability to acquire and apply appropriate knowledge. ***Memory*** indicates the ability to store and retrieve relevant experience according to will.

Learning represents the skill of acquiring knowledge using the method of instruction study.

Experience relates to the understanding that we develop through our past actions. Knowledge can develop over time through successful experience, and experience can lead to expertise. ***Common sense*** refers to the natural and mostly unreflective opinions of humans.

KNOWLEDGE, INTELLIGENCE AND EXPERIENCE

Knowledge is the collection of skills and information a person has acquired through experience. ***Intelligence*** is the ability to apply knowledge.
It also refers to the intellectual or mental capacity of an individual on different grounds, rather than an individual area. Intelligence also includes the capacity for understanding things, self-awareness, applying logic, emotional knowledge, learning, reasoning, problem-solving, creativity, and planning.

Features of Intelligence

- You cannot acquire intelligence after sustained labor.
- It comes as a gift from nature.
- Also, it is not fixed to memory as an intelligent person may have a poor memory.
- Intelligence cannot be a skill which a staffer acquires after some practice.

KNOWLEDGE

Knowledge can also draw its meaning from different lines. Many people have come up with its definitions but all point to the same thing. As an online dictionary will define it, it refers to information, facts, as well as skills that one acquires through either education or experience undertaking a certain task. It is basically the practical or theoretical understanding of a particular subject. Since it is an awareness or familiarity that one gains from experience, it is not inborn or naturally given.

Characteristics of Knowledge

- Knowledge depends and pertains to the circumstances that formed its existence around an idea, statement, or event. Therefore, it is contextual.
- It is only beneficial if applied.
- Its values may change over time.
- It is developed through a learning process.
- Knowledge is dynamic.

- Depends on memory, expertise, exposure, past experiences, opportunities, and the transfer mechanisms.

Difference between Knowledge and Intelligence

- Knowledge is basically the acquisition and acceptance of facts while intelligence points out to the implementation of the facts wisely and perfectly.
- Knowledge can be the same in different individuals, especially if impacted under the same environment, for example, a classroom setup. Intelligence, on the other hand, is unique to individuals. The intelligence of two individuals with similar knowledge will create different results in real life situations.
- Knowledge acquisition is through experience, training, books, practice, materials, and research and is an ever-learning process. On the other hand, intelligence is inborn. It is in one's genes, and the individual is naturally intelligent.
- Intelligence is all about doing the right things the right way without asking for guidance while knowledge needs one to acquire the skills to do something. The latter requires one to ask or learn anything to be able to do it.
- Knowledge helps one know of a problem or situation and how it came to happen. Intelligence, on the other hand, helps an individual to design ways to deal with that situation or solve the problem at hand.
- Knowledge is basically what people or someone already knows. They have been through the process of acquiring it and are passing it on to others. Intelligence is uniquely available to individuals. It helps with the proper implementation of the acquired knowledge.

- Knowledge is primarily a vehicle while intelligence is like the gas that drives it.

Intelligence vs Knowledge

Comparison Table

Characteristics	Intelligence	Knowledge
Acquisition	It is inborn	Acquired through a learning process.
Uniqueness	It is unique to each individual	It is versatile and can be the same with many different people
Nurturing	Cannot be acquired through practice	Requires persistent practice to cultivate
Application	Helps an individual to design ways to deal with that situation or solve the problem at hand.	Helps one know of a problem or situation and how it came to happen.

Difference Between.net

Knowledge Vs Experience

Knowledge: Facts, information, and skills acquired through experience or education; the theoretical or practical understanding of a subject. Experience: The knowledge or skill acquired by a period of practical experience of something especially that gained in a particular profession

Experience is better than knowledge because

1. Experience allows you to understand the theory behind the concepts.
2. Experience allows you to become an expert in your field.
3. Experience is ingrained in your memory.
4. Experience allows you to learn from your mistakes.

COMMON SENSE

Common sense is the knowledge that all humans have. Such knowledge is unspoken and unwritten – we take it for granted. We acquire it imperceptibly from the day we are born. For example, “animals don’t drive cars” or “my mother is older than me”. This knowledge is often used by human experts even when solving very narrow, domain-specific tasks. This common-sense knowledge is something that we learn through experience and curiosity without even being aware of it. We also acquire a great deal of it in our lifetimes.

AI systems do not have common sense knowledge and acquiring it has been seen as important since their beginning. The reason why common sense reasoning was seen as important was that many of these systems were very competent at problem-solving, but they were also brittle, because, they often gave meaningless answers when trying to reason with unusual problem data.

For example, a medical expert might be considering life-threatening surgery on a very elderly patient with a short life expectancy. The expert may rule out the surgery given his or her common sense understanding of the circumstances. An AI expert system, without common sense knowledge, would not have this broader perspective to make such a decision.

COGNITION IN KM

Cognitive Systems can navigate the complexity of human language and can read and understand unstructured information like blogs, tweets, newspapers, videos. They get progressively smarter with every interaction and use.

Cognitive Psychology

- Data, Information and Knowledge
- Kinds of Knowledge
- Expert Knowledge
- Thinking and Learning in Humans

Cognitive psychology tries to identify the cognitive structures and processes that closely relates to skilled performance within an area of operation. It provides a strong background for understanding knowledge and expertise. In general, it is the interdisciplinary study of human intelligence.

The two major components of cognitive psychology are:

[1]Experimental Psychology

This studies the cognitive processes that constitute human intelligence.

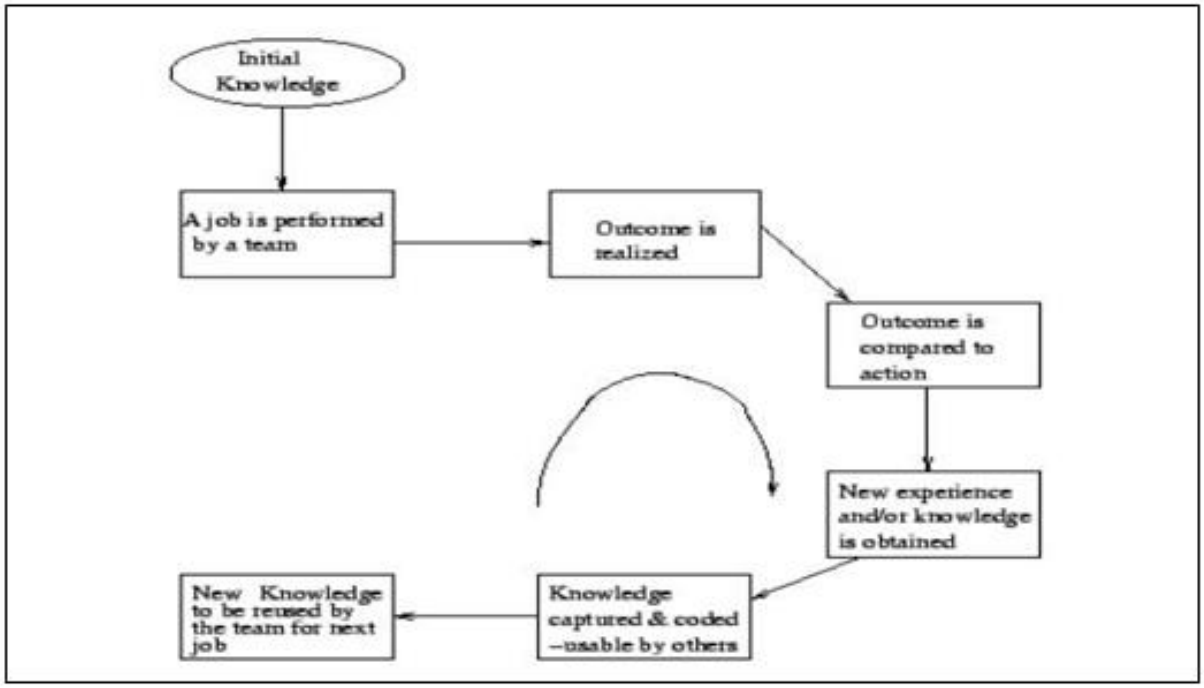
[2]Artificial Intelligence (AI)

This studies the cognition of Computer-based intelligent systems.

The process of eliciting and representing experts knowledge usually involves a knowledge developer and some human experts (domain experts).

In order to gather the knowledge from human experts, the developer usually interviews the experts and asks for information regarding a specific area of expertise. It is almost impossible for humans to provide the completely accurate reports of their mental processes.

The research in the area of cognitive psychology helps to a better understanding of what constitutes knowledge, how knowledge is elicited, and how it should be represented in a corporate knowledge base. Hence, cognitive psychology contributes a great deal to the area of knowledge management.



TYPES OF KNOWLEDGE

[1]Deep Knowledge -Knowledge acquired through years of proper experience

[2]Shallow Knowledge- Minimal understanding of the problem area.

[3]Knowledge as Know-How **-** Accumulated lessons of practical experience.

[4]Reasoning and Heuristics **-** Some of the ways in which humans reason are as follows:

- Reasoning by analogy: This indicates relating one concept to another.
- Formal Reasoning: This indicates reasoning by using deductive (exact) or inductive reasoning.

In case of ***deductive reasoning***, new knowledge is generated by using previously specified knowledge. ***Inductive reasoning*** implies reasoning from a set of facts to a general conclusion. Inductive reasoning is the basis of scientific discovery. A ***case*** is knowledge associated with an operational level.

Common Sense: This implies a type of knowledge that almost every human being possess in varying forms/amounts.

We can also classify knowledge on the basis of whether it is procedural, declarative, semantic, or episodic.

- **Procedural knowledge** represents the understanding of how to carry out a specific procedure.

- **Declarative knowledge** is routine knowledge about which the expert is conscious. It is shallow knowledge that can be readily recalled since it consists of simple and Uncomplicated information. This type of knowledge often resides in short-term memory.
- Semantic knowledge is highly organized, ``chunked'' knowledge that resides mainly in long-term memory. Semantic knowledge can include major concepts, vocabulary, facts, and relationships.
- Episodic knowledge represents the knowledge based on episodes (experimental information). Each episode is usually ``chunked'' in long-term memory.

Another way of classifying knowledge is to find whether it is tacit or explicit.

[5] Tacit knowledge usually gets embedded in human mind through experience.

[6]Explicit knowledge is that which is codified and digitized in documents, books, reports, spreadsheets, memos etc.

EXPERT KNOWLEDGE

It is the information woven inside the mind of an expert for accurately and quickly solving complex problems.

[1] Knowledge chunking -

Knowledge is usually stored in experts long-range memory as chunks. Knowledge chunking helps experts to optimize their memory capacity and enables

them to process the information quickly. ***Chunks*** are groups of ideas that are stored and recalled together as an unit.

[2]Knowledge as an Attribute of Expertise

In most areas of specialization, insight and knowledge accumulate quickly, and the criteria for expert performance usually undergo continuous change. In order to become an expert in a particular area, one is expected to master the necessary knowledge and make significant contributions to the concerned field. The unique performance of a true expert can be easily noticed in the quality of decision making.

The true experts (knowledgeable) are usually found to be more selective about the information they acquire, and also they are better able in acquiring information in a less structured situation. They can quantify soft information, and can categorize problems on the basis of solution procedures that are embedded in the experts long range memory and readily available on recall.

Hence, they tend to use knowledge-based decision strategies starting with known quantities to deduce unknowns. If a first-cut solution path fails, then the expert can trace back a few steps and then proceed again.

Non-Experts

Non-experts use means-end decision strategies to approach the the problem scenario. Non-experts usually focus on goals rather than focusing on essential features of the task which makes the task more time consuming and sometimes unreliable. Specific individuals are found to consistently perform at higher levels than others and they are labeled as experts.

HUMAN THINKING AND LEARNING

Research in the area of artificial intelligence has introduced more structure into human thinking about thinking. Humans do not necessarily receive and process information in exactly the same way as the machines do.

Humans can receive information via seeing, smelling, touching, hearing (sensing) etc., which promotes a way of thinking and learning that is unique to humans.

On macro level, humans and computers can receive inputs from a multitude of sources. Computers can receive inputs from keyboards, touch screens etc.

On micro level, both human brain and CPU of a computer receive information as electrical Impulses The main difference here is that the computers must be programmed to do specific tasks. Performing one task does not necessarily transcend onto other tasks as it may do with humans.

Human learning: Humans learn new facts, integrate them in some way which they think is relevant and organize the result to produce necessary solution, advice and decision. Human learning can occur in the following ways:

- Learning through Experience
- Learning by Example
- Learning by Discovery.